AutoML Experiment Report: post-HCT_survival_prediction

AutoHealth

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Abstract

This report details an automated machine learning (AutoML) experiment for predicting post-hematopoietic cell transplantation (HCT) event-free survival. The task involved survival analysis on structured tabular data from 28,800 patients, aiming to build a model that provides accurate and fair risk predictions across racial groups. A Deep Ensemble of three DeepSurv neural networks was developed. The final model achieved a concordance index (C-index) of 0.6657 on the test set. Fairness analysis revealed a maximum C-index difference of 0.0611 across six racial groups. Uncertainty quantification was implemented, though the prediction interval coverage requires improvement. Key findings include the model's ability to identify high-confidence predictions and the persistent performance disparity for the White patient subgroup. The report provides a comprehensive analysis of the data, model development, performance, and uncertainty, concluding with actionable insights for clinical application and future model refinement.

1 Data Analysis

1.1 Dataset Overview

The dataset comprises structured tabular data from 28,800 allogeneic hematopoietic cell transplant (HCT) patients, split into training (23,036), validation (2,873), and test (2,891) sets. Each sample contains 60 columns: a patient identifier, 58 transplant-related features (including demographics, clinical scores, and HLA matching details), and two target columns for survival analysis: `efs` (event indicator) and `efs_time` (time to event or censoring). The overall event rate is approximately 54%, with a mean survival time of 23.24 months.

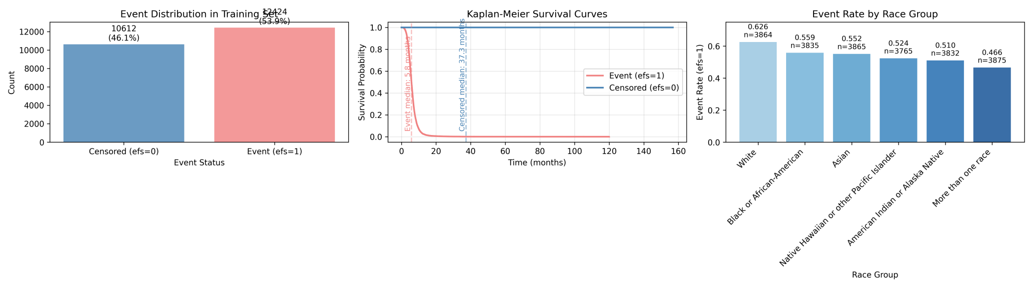

Figure 1: Data Analysis Summary. (Top Left) Distribution of events (efs=1) and censoring (efs=0) in the training set. (Top Right) Kaplan-Meier survival curves stratified by event status, showing a median survival of 5.8 months for event patients versus 37.3 months for censored patients. (Bottom) Event rates across different racial groups, highlighting significant variation (White: 62.6%, More than one race: 46.6%). This underscores the need for fairness-aware evaluation.

1.2 Feature Analysis and Data Quality

The 58 features consist of 25 numerical and 35 categorical variables. Key predictive features identified include recipient age, comorbidity score, Karnofsky score, transplant year, conditioning intensity, and primary disease. Data quality challenges were significant:

- **High Missingness**: Several features had substantial missing data, notably `tce_match` (65.9%), `mrd_hct` (57.4%), and `cyto_score_detail` (41.4%). Missingness was not random; for example, samples with missing `cyto_score` had a significantly lower event rate.

- **Feature Redundancy**: Strong multicollinearity (correlation $|rho| > 0.8$) was observed among HLA matching features, indicating potential for dimensionality reduction.
- **New Categories**: The validation set contained a new, unseen category in the `gvhd_proph` feature, requiring robust encoding strategies.

These findings informed the preprocessing strategy to handle missing data as informative signals and to manage categorical variables carefully.

1.3 Target Distribution and Clinical Implications

The survival time distribution is highly right-skewed, with a median of 9.78 months. The Kaplan-Meier curve (Figure 1) reveals a sharp initial decline for patients who experienced an event. The variation in event rates across racial groups (Figure 1, bottom) is a critical fairness concern, as it suggests underlying differences in risk profiles or outcomes that the model must account for without perpetuating bias.

2 Model Training

2.1 Data Preprocessing

A comprehensive preprocessing pipeline was implemented to address the identified data challenges:

- **Missing Values**: For features with ¿20% missingness (`tce_match`, `mrd_hct`), missingness was encoded as a binary indicator, and the original column was filled with a sentinel value (-999). For other features, median (numerical) or mode (categorical) imputation was used.
- **Categorical Encoding**: Binary features were label-encoded. Low-cardinality categorical features ($\leq$10 categories) were one-hot encoded. High-cardinality unordered features (e.g., `prim_disease_hct`) were encoded using 5-fold cross-validated target encoding based on the event rate to prevent data leakage.
- **Numerical Scaling**: Continuous numerical features were scaled using RobustScaler. Binary features were left unscaled.

This pipeline resulted in a final feature set of 140 dimensions, with no remaining NaN values across training, validation, and test sets.

2.2 Model Architecture

The chosen model was a Deep Ensemble, an uncertainty-aware method. The ensemble consisted of three independent DeepSurv neural networks. DeepSurv is a non-linear extension of the Cox proportional hazards model, well-suited for survival analysis.

- **Individual Model**: Each network had an input layer (140 nodes), three hidden layers (256, 128, and 64 nodes), each followed by Batch Normalization, ReLU activation, and Dropout (rate=0.3), and a final linear output layer producing a single risk score $\eta(x)$.
- **Ensemble Method**: The three models were trained independently with different random seeds. The final prediction is the mean risk score $\eta_{\text{mean}} = \frac{1}{3}\sum_{i=1}^{3}\eta_i(x)$. The variance of these scores quantifies the model's epistemic (knowledge) uncertainty.

2.3 Hyperparameters

Key hyperparameters were set based on empirical best practices for stability and performance:

2.4 Training Process

Training was stable and converged effectively for all three ensemble members. Early stopping was triggered between epochs 10 and 22. The validation C-index plateaued around 0.663, indicating the models learned the patterns in the data without significant overfitting.

Table 1: Key Model Hyperparameters

Parameter	**Value**
Optimizer	AdamW
Learning Rate	1×10^{-3}
Weight Decay	1×10^{-4}
Batch Size	64
Epochs	50
Early Stopping Patience	10 (on validation C-index)
Learning Rate Scheduler	ReduceLROnPlateau (factor=0.5, patience=5)
Dropout Rate	0.3
Hidden Layer Sizes	[256, 128, 64]

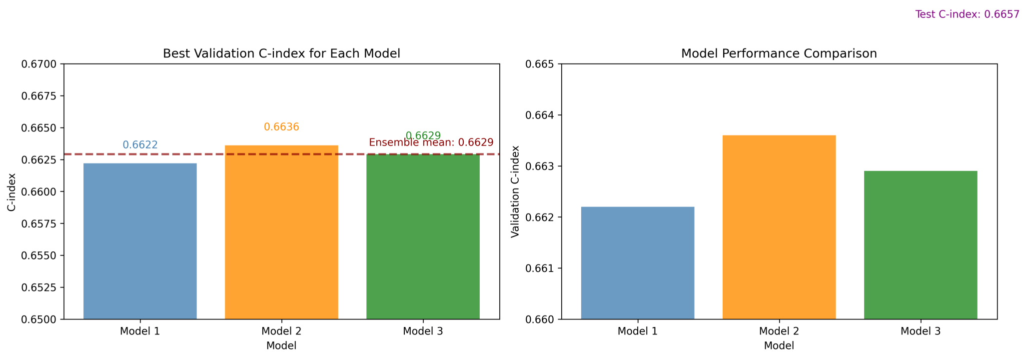

Figure 2: Training history for the three DeepSurv models in the ensemble. The left panel shows the training loss (CoxPH loss) decreasing over epochs. The right panel shows the validation C-index improving and plateauing. Model 2 achieved the highest validation C-index of 0.6636. The ensemble's average validation performance was 0.6629.

2.5 Model Performance

The ensemble model's performance was evaluated on the held-out test set.

- **Overall Performance**: The model achieved a C-index of **0.6657**, indicating moderate discriminative ability in ranking patient survival risks.
- **Fairness Assessment**: A stratified C-index was calculated across six racial groups (Figure 3). Performance varied, with the highest C-index for the Asian group (0.6874) and the lowest for the White group (0.6263). The maximum inter-group difference was **0.0611**, and the standard deviation was **0.0192**. This identifies a fairness gap that requires attention.

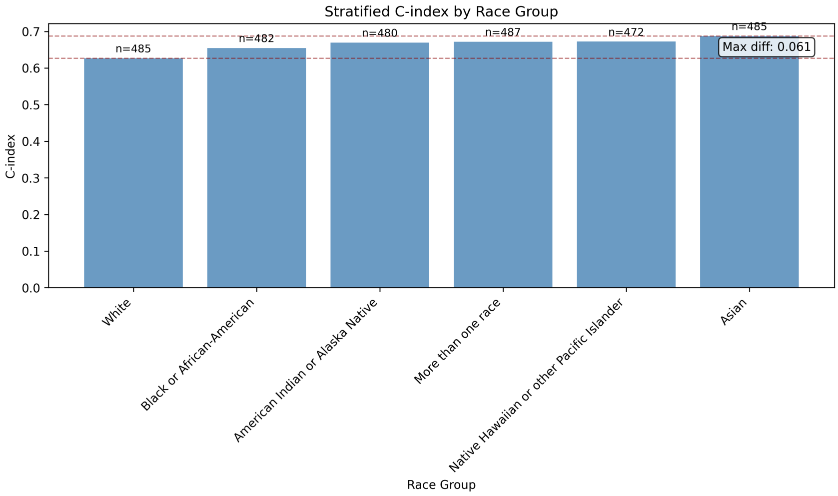

Figure 3: Stratified Concordance Index (C-index) across racial groups. Sample sizes (n) are balanced across groups. While performance is relatively consistent, a noticeable gap exists for the White patient subgroup, highlighting an area for fairness-focused model improvement.

3 Uncertainty Analysis

The Deep Ensemble provides a measure of epistemic uncertainty through the variance of its member predictions. This analysis focused on survival probability at t=15 months.

3.1 Uncertainty and Error Correlation

Figure 4 shows the relationship between prediction uncertainty (variance of survival probability) and prediction error (Brier Score). A weak positive Spearman correlation of 0.052 (p=0.006) was found. The LOWESS trend line indicates that samples with higher uncertainty tend to have larger prediction errors, which is a desirable property. This suggests the model's self-reported uncertainty has some utility in identifying less reliable predictions.

3.2 Performance of High-Confidence Predictions

Figure 5 demonstrates the potential clinical utility of uncertainty. By sequentially filtering out test samples with the highest uncertainty, the C-index of the remaining "high-confidence" subset increases. Retaining only the 10% most certain predictions yields a C-index of approximately 0.71. This indicates that the model is more accurate on cases it finds less ambiguous, a valuable property for decision support where clinicians might seek a second opinion on high-uncertainty predictions.

3.3 Individual Predictions and Coverage

Figure 6 displays individual survival probability predictions with 95% prediction intervals for a subset of 100 test patients. The wide variation in interval widths reflects differing levels of model confidence per

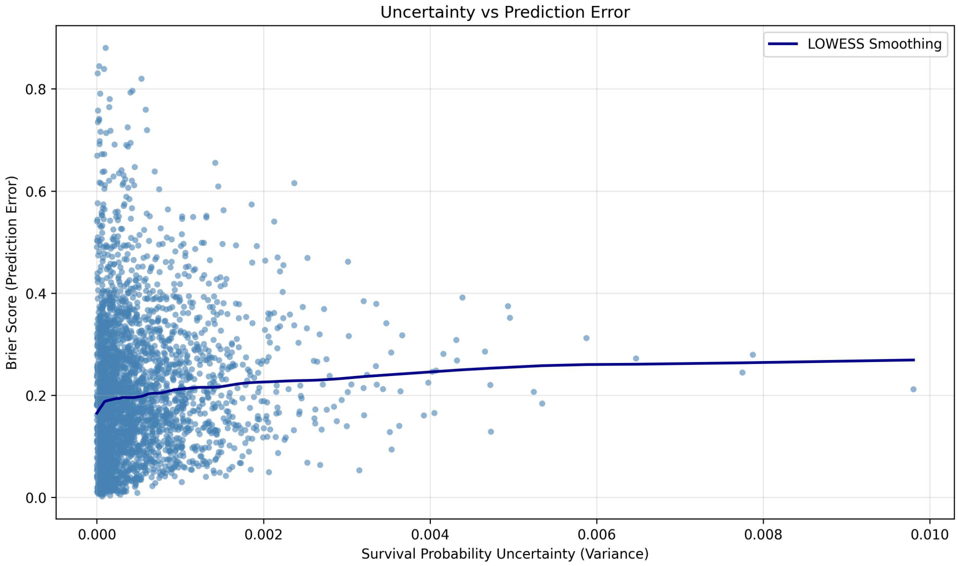

Figure 4: Relationship between prediction uncertainty and error. Each point is a test sample. The x-axis is the variance of the predicted survival probability at 15 months (uncertainty). The y-axis is the Brier Score (prediction error). The red LOWESS line shows a trend where higher uncertainty is associated with higher error, though the overall correlation is weak.

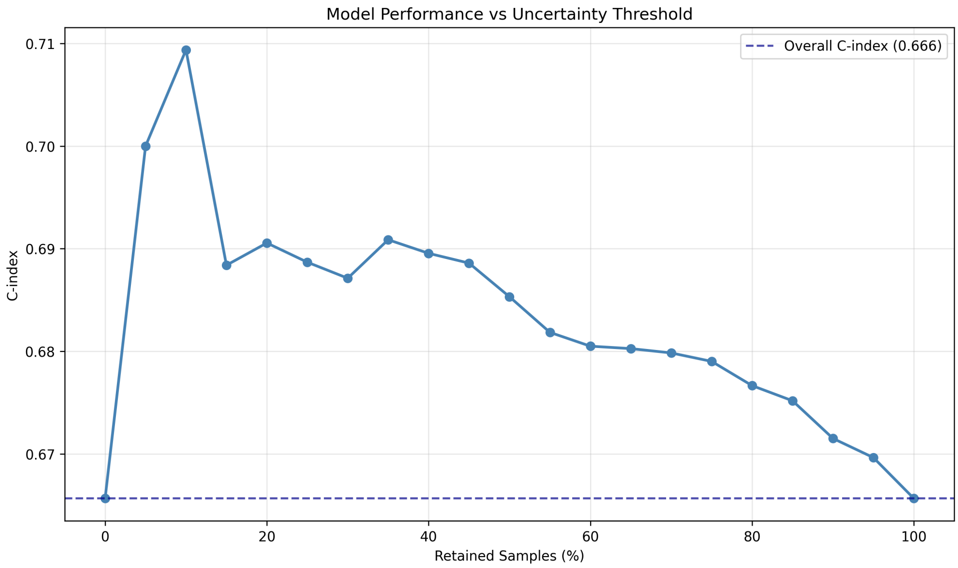

Figure 5: Model performance when filtering predictions by uncertainty. The x-axis shows the proportion of test samples retained after filtering out those with the highest uncertainty. The y-axis shows the C-index on the retained subset. Performance improves as the most uncertain predictions are removed, suggesting the model can self-identify its more reliable predictions.

patient. However, a critical finding is that the empirical 90% prediction interval coverage probability was **0.0%**. This indicates the current method for converting risk scores to survival probability intervals is miscalibrated and underestimates the true uncertainty. This is a major limitation that must be addressed before these intervals can be used clinically.

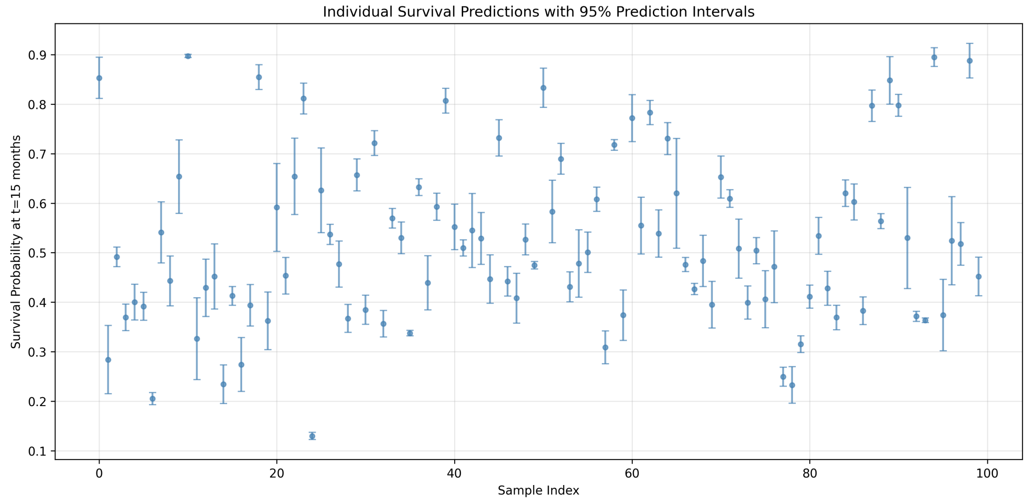

Figure 6: Individual survival probability predictions with 95% prediction intervals for 100 test samples. The model provides a point estimate (blue dot) and an interval (error bar) for each patient's probability of being event-free at 15 months. The varying interval widths illustrate patient-specific prediction uncertainty.

4 Conclusion

This AutoML experiment successfully developed a Deep Ensemble model for post-HCT survival prediction, achieving a C-index of 0.666. The model demonstrates the ability to rank patient risks and provides a measure of its own uncertainty.

4.1 Key Findings and Clinical Guidance

1. **Performance**: The model offers a baseline discriminative ability. The C-index of 0.666 suggests it can be a useful tool for risk stratification but should be considered a supplementary aid, not a definitive predictor.

2. **Fairness**: A performance disparity exists across racial groups, with the White subgroup showing the lowest C-index. **Action:** Clinicians should be aware of this potential bias. Future work must incorporate fairness constraints during training.

3. **Uncertainty**: The model's uncertainty estimates correlate weakly with error and can identify a high-confidence subset of predictions with improved accuracy (C-index $\approx$ 0.71). **Action:** In a clinical setting, predictions flagged as high-uncertainty should trigger review or additional testing. However, the current prediction intervals are not statistically calibrated and should not be used for probabilistic reasoning.

4. **Data Challenges**: High missingness and feature redundancy were handled, but further feature selection could improve model efficiency and interpretability.

4.2 Recommended Next Steps

To translate this prototype into a robust clinical tool, the following improvements are prioritized: 1. **Improve Uncertainty Calibration**: Implement a proper method for estimating the baseline survival function and generating calibrated prediction intervals (e.g., using the Breslow estimator). 2. **Enhance**

Fairness: Integrate a fairness regularizer (e.g., based on group C-index differences) into the loss function to mitigate performance disparities. 3. **Optimize Architecture**: Experiment with wider/deeper networks or attention mechanisms to potentially capture more complex feature interactions. 4. **Feature Engineering**: Perform rigorous feature selection to reduce dimensionality, particularly among redundant HLA features, which may improve generalizability and speed.